Probability
Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 From a group of 5 red and 5 black cards, one card is selected at random then replaced and a second card selected. What is the probability that the cards chosen are the same colour? 1

½ BB + RR = 1/2

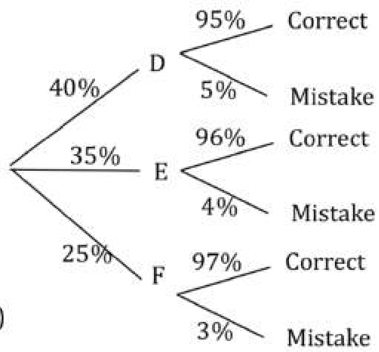
2 Using the following sets: $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 4, 5, 6, 7\}$. List the elements of:

a. $A \cap B$ $\{3, 4, 5\}$ 1

b. $A \cup B$ $\{1, 2, 3, 4, 5, 6, 7\}$ 1

3 Three employees, David, Eliza and Frank, work for the same company. They respectively handle 40%, 35% and 25% of the company's orders. It has been found that they make mistakes in 5%, 4% and 3% of their orders respectively.

a. Use a tree diagram to display this information. 2

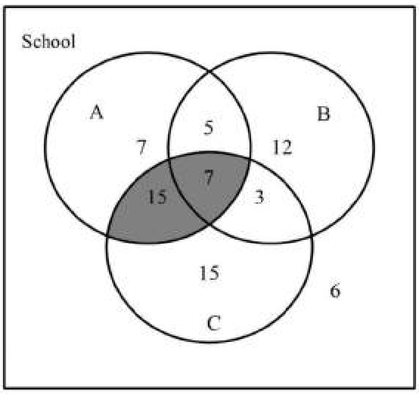


b. Calculate the probability that an order handled by the company has a mistake. 2

$$\begin{aligned} P(\text{Mistake}) &= P(\text{D Mistake}) + P(\text{E Mistake}) + P(\text{F Mistake}) \\ &= 0.40 \times 0.05 + 0.35 \times 0.04 + 0.25 \times 0.03 \\ &= 0.0415 \text{ (or } 4.15\%) \end{aligned}$$

$\therefore$ Probability of a mistake is 4.15%

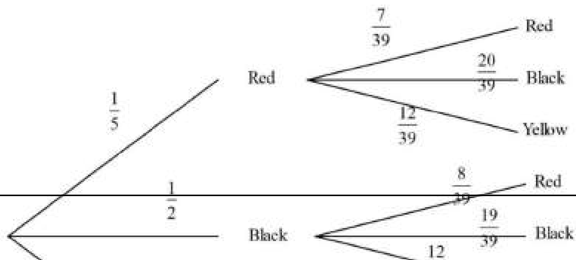
4 The Venn diagram shows the membership by the 70 students in a school, in three groups, the archery club (A), the basketball teams (B) and the choir (C). A student from the school is chosen at random. What is the probability that the student is neither a choir member nor in the archery club? 2



$$P(A \cap C) = \frac{15 + 7}{70} = \frac{22}{70}$$

$$P(\text{ not}(A \cap C)) = 1 - \frac{22}{70} = \frac{48}{70} = \frac{24}{35}$$

5 Mike draws two marbles, with no replacement, from a barrel. The barrel contains 8 red marbles,



20 black marbles and 12 yellow marbles. A probability tree diagram has been started to model this situation.

a. Complete the tree by writing the remaining probabilities on the branches. 1

b. What is the probability that exactly one of the two marbles are yellow? 1

$$P(\text{ 1 is Yellow }) = P(\text{RY}) + P(\text{BY}) + P(\text{YR}) + P(\text{YB})$$
$$= \frac{1}{5} \times \frac{12}{39} + \frac{1}{2} \times \frac{12}{39} + \frac{3}{10} \times \frac{8}{39} + \frac{3}{10} \times \frac{20}{39} = \frac{28}{65}$$

c. What is the probability that the two marbles are not the same colour? 1

$$P(\text{ Same colour }) = P(R\,R) + P(\text{BB}) + P(\text{YY})$$
$$= \frac{1}{5} \times \frac{7}{39} + \frac{1}{2} \times \frac{19}{39} + \frac{3}{10} \times \frac{11}{39} = \frac{71}{195}$$
$$P(\text{ Not Same colour }) = 1 - \frac{71}{195} = \frac{124}{195}$$

6 Vanessa collects data on various aspects of life for the 30 students in her class. Some of the results are summarised in the table below. One student is chosen at random from the class.

Aspect	Frequency
Has red hair	10
Plays sport	17
Has a job	13
Has red hair and plays sport	6
Has red hair and has a job	5
Has a job and plays sport	9
Has red hair, plays sport and has a job	3
Does not have any of the characteristics above	7

a. What is the probability that the student plays sport and has red hair? 1

b. What is the probability that the student plays sport or has red hair, or both? 1

c. What is the probability that the student does not have a job nor do they have red hair? 1

a)
$$P(R \cap S) = \frac{6}{30} = \frac{1}{5}$$

b)
$$P(R \cup S) = P(A) + P(B) - P(A \cap B)$$
$$= \frac{10}{30} + \frac{17}{30} - \frac{6}{30} = \frac{21}{30} = \frac{7}{10}$$

c)
$$P(R \cup J) = P(R) + P(J) - P(R \cap J)$$
$$= \frac{10}{30} + \frac{13}{30} - \frac{5}{30} = \frac{18}{30} = \frac{3}{5}$$
$$\overline{P(R \cup J)} = 1 - \frac{3}{5} = \frac{2}{5}$$

7 Paul buys three tickets in a raffle which has three equal prizes. There are 100 tickets sold in the raffle.

a. What is the probability that he wins all three prizes? 1

b. If he does not win the first prize which is drawn, what is the probability that he wins the next two prizes? 1

a)
$$P(WWW) = \frac{3}{100} \times \frac{2}{99} \times \frac{1}{98} = \frac{1}{161\,700}$$

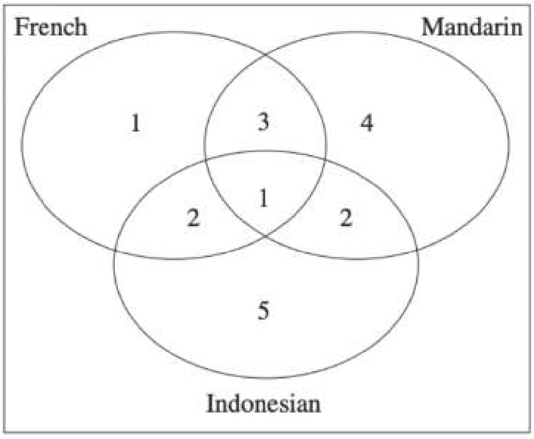
b) If doesn't win first, he still has 3 tickets from 99, since one drawn

$$P(\text{Wins next 2 given doesn't win first}) = P(WW \text{ after } L)$$
$$= \frac{3}{\quad} \times \frac{2}{\quad} = \frac{1}{\quad}$$

8 Ben is playing basketball. Th probability that he misses two free throws successively? 1

$$0.3 \times 0.3 = 0.09$$

9 In a mixed language class, students study French, Mandarin and Indonesian. The number of students who study each language are shown in the Venn diagram. A student who studies Indonesian is selected at random. What is the probability that he/she also studies French? 1



10 Indonesian 3 both French and Indonesian

3/10

10 In a mathematics class, 15 students passed a test and 10 students did not pass the test. Among those who passed the test, 12 had completed their homework and among those who did not pass the test, 2 had completed their homework. A student who had completed the homework is randomly selected from this class. Find the probability that this student passed the test. 4

T – pass the test, H – completed the homework

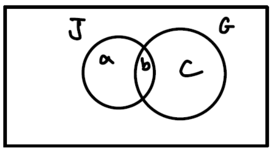
$$P(T|H) = \frac{P(T \cap H)}{P(H)} = \frac{\frac{12}{25}}{\frac{14}{25}} = \frac{12}{14} = \frac{6}{7}$$

11 On a given day, the probability that it is sunny is 0.3, the probability that it is windy is 0.6, and the probability that it is both sunny and windy is 0.15. What is the probability that it is sunny or windy (or both)? 2

$P(S \text{ or } W) = P(S) + P(W) - P(S \text{ and } W)$

$= 0.3 + 0.6 - 0.15 = 0.75$

12 All of Ben's friends either jog or go to the gym. He knows that 60% of his friends' jog, and 30% jog and go to the gym. Show that 40% of Ben's friends go to the gym, but do not jog? 2



$a + b + c = 1$ — ①

$a + b = 0.6$ — ②

$b = 0.3$ — ③

sub ③ into ②:

$a + 0.3 = 0.6$

$a = 0.3$

sub $a = 0.3$ into ②

$a + 0.3 = 0.6$

$a = 0.3$

sub $a = b = 0.3$ into ①:

$0.3 + 0.3 + c = 1$

$c = 1 - 0.6 = 0.4$

∴ 40% go to the gym but do not jog.

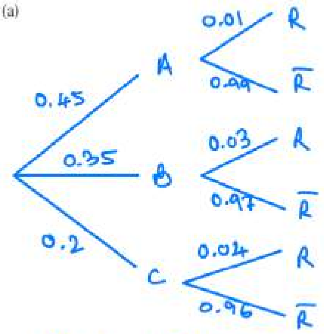
13 A supermarket buys 45% of its fruit from farm A, 35% of its fruit from farm B and the rest from farm C. The probability that a randomly chosen fruit supplied by farms A, B or C is rotten is 0.01, 0.03 and 0.04 respectively.

a. Draw a probability tree diagram representing this information. 1

b. Find the probability that a randomly chosen fruit is rotten. 1

c. Given that a randomly chosen fruit is NOT rotten, what is the probability that it was supplied by farm B? 1

(a)



(b) $P(R) = 0.45 \times 0.01 + 0.35 \times 0.03 + 0.2 \times 0.04 = 0.023$

(c) $P(B \mid \bar{R}) = \dfrac{P(B \text{ and } \bar{R})}{P(\bar{R})}$

$= \dfrac{0.35 \times 0.97}{1 - 0.023}$

$= \dfrac{679}{1954}$ or 0.3475 (4 d.p)

Probability
Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 The probability that it rains on Saturday is $\frac{1}{3}$. The probability that it rains on Sunday is $\frac{2}{5}$. What is the probability that it does not rain at all during the weekend? 1

Probability not raining on Saturday: $1-\frac{1}{3}=\frac{2}{3}$

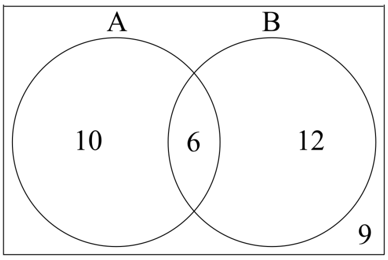
Probability not raining on Sunday: $1-\frac{2}{5}=\frac{3}{5}$

Probability not raining on Saturday and not raining on Sunday:

$\frac{2}{3}\times\frac{3}{5}=\frac{6}{15}=\frac{2}{5}$ (multiplication law)

2 Consider the Venn diagram. According to the diagram, what is $P(B)$? 1

$$P\,(B)=\frac{P(A\cap B)}{P(B)}=\frac{6}{18}=\frac{1}{3}$$



3 A survey of a Year 11 class indicates that $\frac{1}{9}$ of the students play soccer, $\frac{1}{5}$ of the students are in the chess club, and $\frac{1}{12}$ of the students both play soccer and are in the chess club. If a student is chosen at random, what is the probability that they play soccer or are in the chess club (or both)? 2

$P\,(A\cup B)=P\,(A)+P\,(B)-P(A\cap B)$ (see Reference Sheet)

$P\,(soccer\ or\ chess\ club)=\frac{1}{9}+\frac{1}{5}-\frac{1}{12}=\frac{41}{180}$

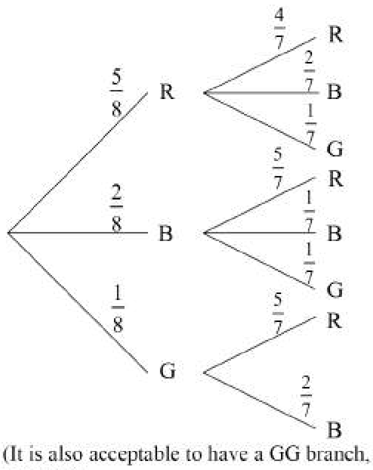
4 In a bag there are 5 red marbles, 2 blue marbles and 1 green marble. One marble is selected randomly and hidden from view. A second marble is then selected and hidden from view.

a. Draw a tree diagram to show all of the possible outcomes and the associated probability of each branch. 2

b. What is the probability that neither of the chosen marbles is blue? 1

c. One of the marbles is revealed to be blue. What is the probability that the other marble is blue? 1

(a)



(It is also acceptable to have a GG branch, with the probability for a second green being 0.)

(b) Probability neither is blue is the same as the probability as getting two reds or a green and a red:

$P\,(\overline{BB})=P\,(RR\cup RG\cup GR)$

$=\frac{5}{8}\times\frac{4}{7}+\frac{5}{8}\times\frac{1}{7}+\frac{1}{8}\times\frac{5}{7}=\frac{15}{28}$

(c) $P\,(B)=\frac{1}{7}$

(This can be seen as the probability of the second blue branch coming off of the first blue branch.)

5 The sample space when a fair die is rolled is $\{1, 2, 3, 4, 5, 6\}$. Each outcome is equally likely. For which pair of events, A and B, are A and B independent? 1

A. $A = \{2, 3\}$ and $B = \{2, 3, 5, 6\}$

B. $A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$

C. $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$

D. $A = \{1, 3, 5\}$ and $B = \{3, 6\}$

A and B independent not to be confused with mutually exclusive.
P(A and B) = P(A) x P(B)

$A: \frac{2}{6} \neq \frac{2}{6} \times \frac{4}{6}$ $C: \frac{1}{6} \neq \frac{1}{2} \times \frac{1}{2}$

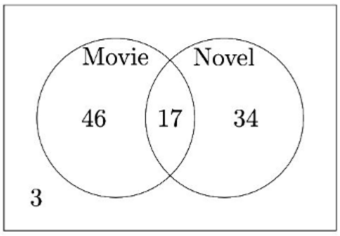
$B: 0 \neq \frac{1}{2} \times \frac{1}{2}$ $D: \frac{1}{6} = \frac{3}{6} \times \frac{2}{6}$

D

6 An integer n is chosen at random from the set $\{5, 7, 9, 11\}$. An integer p is also chosen at random from the set $\{2, 6, 10, 14, 18\}$. What is the probability that $n + p = 23$? 1

A. 0.1 B. 0.2 C. 2.5 D. 0.3

7 A magazine surveyed 100 readers about whether they had read a certain novel or watched its movies adaptation. The data collected is represented in the Venn diagram below. 1

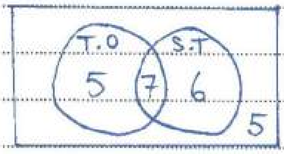


$\frac{17}{63}$

A reader was selected at random. If the reader selected has watched the movie, what is the probability that they had also read the novel?

8 In a class of 23 students, on Sunday night 12 watched The Office, 13 watched Stranger Things while 7 watched both.

a. Find the probability that a student chosen at random watched neither. 1



$P(\text{neither}) = \frac{5}{23}$

b. Find the probability that they watched The Office, given that they watched Stranger Things. 1

$P(\text{The Office} \setminus \text{Stranger Things}) = \frac{7}{13}$

9 Two events, A and B, are such that $P(A) = \frac{3}{5}$ and $P(B) = \frac{1}{4}$. If A' denotes the complement of A, calculate $P(A' \cap B)$ when:

i. $P(A \cup B) = \frac{3}{4}$ $P(A' \cap B) = \frac{3}{20}$ 1

ii. A and B are mutually exclusive. $P(A' \cap B) = \frac{5}{20}$ 1

10 For events A and B, $P(A|B) = \frac{2}{5}$ and $P(B|A) = \frac{1}{3}$. Let $P(A \cap B) = p$. What is the value of $P(A)$? 2

$P(B|A) = \frac{1}{3}$; $P(A \cap B) = p$

$P(B|A) = \frac{P(B \cap A)}{P(A)}$

$\frac{1}{3} = \frac{p}{P(A)} \longrightarrow \therefore P(A) = 3p$

11 The probability that a certain girl wakes up late in the morning is 0.6. When she wakes up late, the probability that she is late to work is 0.85. When she wakes up on time, the probability that she is late to work is 0.4.

a. What is the probability that she is late to work on any morning? 2

P(late) = P(wake up late and late to work) +
P(wake up on time and late to work)

$P(\text{late}) = 0.6 \times 0.85 + 0.4 \times 0.4 = 0.67$

b. Upon seeing the girl arriving late, her boss asks her if she overslept. What is the probability that he truthful answer is 'yes'? 2

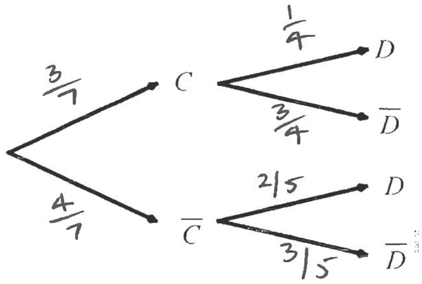
$$P(\text{ wakes up late}|\text{ arrives late}) = \frac{P(\text{wakes up late} \cap \text{arrive late})}{P(\text{late})}$$
$$= \frac{0.6 \times 0.85}{0.67} = \frac{51}{67}$$

12 A pet ownership survey resulted in the following results:

$$P(C) = \frac{3}{7}, P(D|\bar{C}) = \frac{2}{5} \text{ and } P(\bar{D}|C) = \frac{3}{4}$$

Where C is the event that "a person has a cat" and D is the even that "a person has a dog".

a. Complete the probability tree by marking a probability on each branch. 2



b. If one person is chosen at random, find the probability that the person has:

c. i. a cat and a dog 1

$P(C \cap D) = \frac{3}{7} \times \frac{1}{4} = \frac{3}{28}$

ii. at least one pet (cat or dog) 2

$P(\text{at least 1 pet}) = 1 - P(\bar{D}\bar{C})$
$= 1 - \frac{4}{7} \times \frac{3}{5} = \frac{23}{35}$

13 For events A and B from a sample space, $P(A|B) = \frac{3}{4}$ and $P(B) = \frac{1}{3}$.

a. Calculate $P(A \cap B)$ $P(A \cap B) = P(A|B) \cdot P(B) = \frac{3}{4} \cdot \frac{1}{3} = \frac{1}{4}$ 1

b. Calculuate $P(\bar{A} \cap B)$ where $\bar{A}$ denotes the complement of A. 1

$P(\bar{A} \cap B) = P(\bar{A}|B) \cdot P(B) = \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{12}$

c. If A and B are independent, calculuate $P(A \cup B)$. 2

"Independent" means $P(A) = P(A|B) = \frac{3}{4}$

$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{3}{4} + \frac{1}{3} - \frac{1}{4} = \frac{5}{6}$

Probability
Topic Test 3 Solution

Total Marks: 30

Time: 50 minutes

1 It is known for a large population that at the beginning of winter, 15% of people will be infected with a particular virus.

a. Four people are selected at random. Find the probability that at least one of them has the virus. 1

$P(\text{at least one has virus}) = 1 - P(\text{none have virus})$

$= 1 - P(\bar{V}\bar{V}\bar{V}\bar{V})$

$= 1 - (0.85)^4$ — 1

$= 0.47799375$ — 1

b. What is the smallest number of people a drug company would need to test to have a greater than 95% chance that at least one of the tested people has the virus? 3

let n be the least number of people

$P(\text{at least one of } n \text{ has the virus}) > 0.95$

$1 - (0.85)^n > 0.95$ — 1

$-(0.85)^n > -0.05$

$(0.85)^n < 0.05$

$\ln (0.85)^n < \ln 0.05$

$n \ln (0.85) < \ln 0.05$

$n > \dfrac{\ln 0.05}{\ln 0.85}$ — 1

$n > 18.43312827$

$\therefore$ 19 people would need to be tested. — 1

c. As winter progresses the virus spreads further so the health authorities decide to trial a new medication to try and stop the spread of the virus. The two-way table shows the number of people in a trial. (NOTE: the trial consists of those talking the medication and those in a control group).

	Taking Medication	Control Group	
Virus	204	205	409
No Virus	212	209	421
	416	414	830

i. What percentage of people in the trial has the virus? 1

$\dfrac{204 + 205}{830} = \dfrac{409}{830} = 0.4927710843 = 49.27710843\ \%$

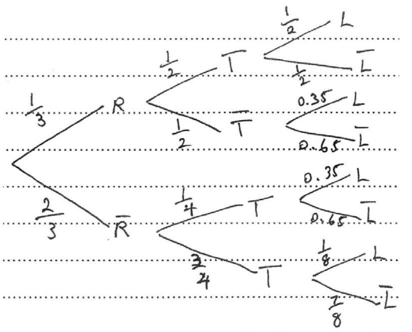
ii. What percentage of people in the control group had the virus? 1

$\dfrac{205}{205+209} = \dfrac{205}{414} = 0.495169082 = 49.5169082\ \%$

iii. Giving a reason, determine if it is worth the health authorities using this new medication. 2

Not worth using the new medication since the "taking medication" group $\left(\frac{204}{416} = 49.03846154\%\right)$ and the "control" group $\left(\frac{205}{414} = 49.51690821\%\right)$ had a similar incidence (occurrence) of the virus.

2 Neville lives in a town where one third of the days are rainy days. If it is a rainy day, the chance of heavy traffic is $\frac{1}{2}$. If it is a not a rainy day, the chance of heavy traffic is $\frac{1}{4}$. If it is a rainy day and there is heavy traffic, there is a 50% chance that Neville will arrive late to work. On the other hand, the probability of being late is reduced to $\frac{1}{8}$ if it is not a rainy day and there is no heavy traffic. In other situations, the probability of Neville being late to work is 0.35. On a random day, Neville arrives late to work. What is the probability that it is a rainy day? 2



$$P(R/L) = \frac{P(L \cap R)}{P(L)}$$

$$= \frac{\frac{1}{3}\times\frac{1}{2}\times\frac{1}{2} + \frac{1}{3}\times\frac{1}{2}\times\frac{35}{100}}{\frac{1}{3}\times\frac{1}{2}\times\frac{1}{2} + \frac{1}{3}\times\frac{1}{2}\times\frac{35}{100} + \frac{2}{3}\times\frac{1}{4}\times\frac{35}{100} + \frac{2}{3}\times\frac{3}{4}\times}$$

$$= \frac{34}{63}$$

3 Data from 200 pupils is tabulated below. The data gathered indicates which hand they write with and their favourite colour.

A pupil is selected at random.

	Green	Red	Blue	Total
Left	17	8	21	46
Right	44	71	39	154
Total	61	79	60	200

a. Find the probability that the pupil is right-handed. $\frac{154}{200} = \frac{77}{100}$ or 0.77 1

b. Find the probability that the pupil prefers green, given that they are left-handed. 1

$P(\text{green}|\text{left}) = \frac{17}{46}$

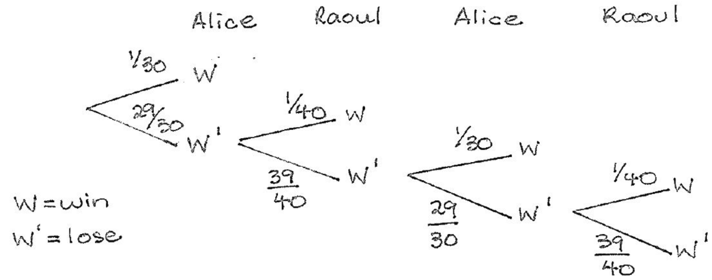
c. Are the events of being left-handed and preferring green independent? Justify your answer with a calculation. 1

$P(\text{green}) = \frac{61}{200}$ ✓ $P(\text{green}|\text{left}) = \frac{17}{46}$

Since these differ the events are dependent

4 Alice and Raoul take turns throwing darts at a dartboard. The winner is whoever hits the bullseye first. Alice has a $\frac{1}{30}$ chance of hitting the bullseye, while Raoul has a $\frac{1}{40}$ chance of hitting the bullseye. Alice throws the first dart.

a. Draw a tree diagram for the first four throws of the games (two throws for Alice and two for Raoul). 1



b. What is the probability that Alice wins on her first or second throw? 1

$$P\left(\begin{array}{c}\text{Alice wins on}\\ \text{1st or 2nd}\\ \text{throw}\end{array}\right) = \frac{1}{30} + \left(\frac{29}{30}\right)\left(\frac{39}{40}\right)\left(\frac{1}{30}\right) = \frac{259}{4000}$$

c. What is the probability that Alice will eventually win the game? (This question requires the knowledge of Yr12 Series & Sequences) 3

P(Alice wins game)

$$= \frac{1}{30} + \left(\frac{29}{30}\right)\left(\frac{39}{40}\right)\left(\frac{1}{30}\right) + \left(\frac{29}{30}\right)\left(\frac{39}{40}\right)\left(\frac{29}{30}\right)\left(\frac{39}{40}\right)\left(\frac{1}{30}\right) + \cdots$$

geometric series $a = \frac{1}{30}$, $r = \left(\frac{29}{30}\right)\left(\frac{39}{40}\right)$

S_∞ exists as $|r| < 1$

$$= \frac{a}{1-r} = \frac{\frac{1}{30}}{1 - \left(\frac{29}{30}\right)\left(\frac{39}{40}\right)} = \frac{40}{69}$$

5 An ice cream company launches a promotion in which customers have a chance of winning a limited-edition item. The probability of winning this prize is 0.07 with each individual purchase. Each person can only win the prize once. Benjamin decides to purchase one ice block each day until he wins the prize.

a. Find the probability that he wins the item on the first or second day, correct to two decimal places. 2

$P(\overline{W}) = 0.93$

$P(A) = P(W) + P(\overline{W}W) = 0.07 + 0.93 \times 0.07 = 0.1351$

Hence the probability that Benjamin wins on the first or second day is 0.14.

b. Hence find the smallest number of days for which the probability that Benjamin will win the limited-edition item, is greater than 80%. 2

Let B be the event that Benjamin wins eventually.

The sequence of probabilities that Benjamin wins on a certain day:

$0.07, 0.93 \times 0.07, 0.93^2 \times 0.07, \cdots$

Forms a geometric sequence:

$$a = 0.07\ ,\ r = 0.93$$

$$P(B) > 0.8$$

$$\frac{0.07(1-0.93^n)}{1-0.93} > 0.8$$

$$1 - 0.93^n > 0.8$$

$$-0.93^n > -0.2$$

$$0.93^n < 0.2$$

$$n > \log_0 .930.2$$

$$n > \frac{\log_1 00.2}{\log_1 00.93}$$

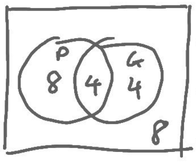
$$n > 22.18$$

Hence the smallest number of days is 23.

6 A teacher in college asks her mathematics students what other subjects they are studying. She finds that, of her 24 students, 12 study physics, 8 study geography and 4 study both.

a. A student is chosen at random from the class. Determine whether the event 'the student studies physics and the event 'the student studies geography' are independent. 2

$P(P \cap G) = P(P) \times P(G)$



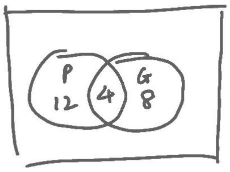
$LHS = \frac{4}{24} = \frac{1}{6}$ OR $LHS = \frac{4}{24} = \frac{1}{6}$

$RHS = \frac{12}{24} \times \frac{8}{24}$ $RHS = \frac{16}{24} \times \frac{12}{24}$

$= \frac{1}{2} \times \frac{1}{3}$ $= \frac{2}{3} \times \frac{1}{2}$

$= \frac{1}{6}$ $= \frac{1}{3}$

$= LHS$ ∴ independent $\neq LHS$ ∴ not independent



b. It is known that for the whole college: 2

- The probability of a student studying mathematics is $\frac{1}{5}$.
- The probability of a student studying biology is $\frac{1}{6}$.
- The probability of a student studying biology given that they study mathematics is $\frac{3}{8}$.

Calculate the probability that in the whole college a student studies mathematics or biology or both.

$P(M) = \frac{1}{5}$, $P(B) = \frac{1}{6}$, $P(B|M) = \frac{3}{8}$ $P(B \cup M) = ?$

$P(B|M) = \frac{P(B \cap M)}{P(M)}$

$\frac{3}{8} = \frac{P(B \cap M)}{\frac{1}{5}}$

$P(B \cap M) = \frac{3}{40}$

$P(B \cup M) = P(M) + P(B) - P(B \cap M)$

$= \frac{1}{5} + \frac{1}{6} - \frac{3}{40} = \frac{7}{24}$

7 One halt percent (0.5%) of a country has a certain viral disease. A test is developed for the disease. The test gives a false positive 3% of the time, and a false negative 2% of the time.

a. Show that the probability that Andy, a randomly selected person, tests positive is 0.03475. 2

$P(T) = P(T/D) \cdot P(D) + P(T/\bar{D}) \cdot P(\bar{D})$

$= (0.98)(0.005) + (0.03)(0.995)$

$= 0.03475$

[*Hint:* in this question, let D be the event that Andy has the disease, and $\bar{D}$ be the event Andy does not have it. Let T be the event that Andy's test comes back positive.] 2

b. Andy just got the bad news that his test came back positive. Find the probability that Andy actually has the disease.

$P(D|T) = \frac{P(D \cap T)}{P(T)} = \frac{P(T \cap D)}{P(T)}$

$= \frac{P(T|D) \cdot P(D)}{P(T)} = \frac{(0.98)(0.005)}{0.03475} = 0.141 \doteq 14\%$